

Lesson 2.18: Buying Low, Rebalancing, and Return Math

Unit: Unit 2, Me, My Goals, My Money (Individual Growth and Life Readiness) **Topic:** 2.4 The Gilded Age Market Investor Challenge **Grade:** 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 1 class period of 40 minutes **Room:** classroom with a board (the market is a paper game; see Project 02)

Standards

- NYS module line: NYS IGLR 4. Financial Readiness e) Determine ways for saving money
- CTE theme line: CTE FCL 2. Saving, Investing, and Sharing c) Explore different savings and investment tools and methods
- CTE theme line: CTE FCL 2. Saving, Investing, and Sharing f) Apply decision-making skills to saving, investing, and donating decisions
- NYS FACS Learning Standard 3 (Intermediate), Key Idea: Students will understand and be able to manage personal resources of talent, time, energy, and money and make effective decisions in order to balance their obligations to work, family, and self.; Performance Indicator: Students understand the resources available to them, make informed decisions about the use of those resources, and know some ways to expand resources.
- CDOS Standard 3a Universal Foundation Skills: Basic Skills
- National FCS 3.0: 3.3.4 Compare different ways to save and invest.

Enduring understanding and essential question

Big idea: A number by itself (up \$400, down \$600) does not tell you how well you did. Percent return does, because it compares what you ended with to what you started with, and it lets you compare a safe ledger to a risky one across the same rounds. **Essential question:** At the bottom, when everyone is selling and nobody is buying, is the cheap share a bargain or a trap, and how would the math help you decide?

Objectives

By the end of this lesson, students will be able to:

1. (Low) Given a starting value and an ending value, state the percent return formula (ending minus starting, divided by starting, times 100) and compute it for one example.
2. (Mid) Given their own ledger, compute their total's percent return from the \$1,000 they started with in 1870, and the percent change of one company from the 1928 peak to the 1932 bottom.
3. (High) Given a safe ledger, a one-company ledger and a mixed ledger tracked over the same rounds, compare their percent returns at the peak (1928), in the crash (1929) and at the bottom (1932, today), and decide in writing which they would rather have held and why.

Vocabulary

Tier 2 (general academic): compare, percent, change, recover Tier 3 (FACS): percent return, buying low, rebalancing, peak, bottom, starting value, ending value. See the unit vocabulary list for the Spanish column.

Materials and setup

- Handout 02.14 - The Gilded Age Market Kit.md (teacher): page 2 for the Round 8 prices, page 3 for the Round 8 dispatch
- Handout 02.19 - Paper Ledger.md (student), in every folder
- Handout 02.18 - Return Math.md, one per student
- Calculators, one per student
- Investor Journal in the folder
- Slides: Slides 02.18 - Buying Low Rebalancing and Return Math
- Exit cards
- Setup notes: before class, write “Round 8, 1932” and the Round 8 prices on the board (RAIL 10, OIL 25, GAS 20, ELEC 10, STEEL 25, AUTO 8, RADIO 3) and cover them until Step 1, because the do now works better if students have not seen them yet. Have the Round 6 prices still visible somewhere (the peak) for Part B. Tell the math teacher this is percent change week. The market freezes at Round 8 at the end of this period; collect the ledgers at the bell if you want to build the leaderboard yourself before Lesson 2.19.

Pre-assessment

The grade 8 stretch students already computed a percent drop in Lesson 2.17. Everyone else has subtracted from \$1,000 in the journal for seven rounds but not divided. The do now shows who can turn “up \$20 a share” into a percent without help; expect few.

Do now and hook (Time: 3 min)

On the board (with the Round 8 prices still covered): “In 1928, GAS was \$40 a share. Suppose it fell to \$20 and you bought 10 shares for \$200. If the price ever climbs back to \$40, how much do you make? Now the harder one: what percent of your money did you make?”

Debrief: \$20 a share, \$200 on 10 shares. The percent is 200 divided by 200, which is 100 percent: you doubled your money. Then the hook: “That is what buying low means. It is also what everyone who sells at the bottom hands to the person who buys from them. Today you learn the one formula that tells you who won.” Then uncover the board.

Procedure

Step	Teacher will	Students will	Time
1. Round 8 opens: the bottom	Uncover the Round 8 prices. Give the room a minute to write them in the ledger and work out the Round 8 total	Fill the Round 8 price column and the Round 8 holdings table. Listen. Answer the question.	5 min

	(shares times price, plus cash, plus the bank, which just paid 3 percent again). Then read the Round 8 dispatch aloud (kit page 3): prices have fallen for almost three years; on July 8, 1932, the Dow Jones average closed at 41, eighty nine percent below its 1929 peak; about one worker in four has no job; thousands of banks have failed and the money in them is gone (the game's bank survives; most did not); General Motors sells for what a good shirt costs. Put the kit's question to the room and take two answers: "Everyone is selling and nobody is buying. If you have cash, what does that make this?" Do not open trading yet.		
2. Direct instruction and Return Math, Parts A and B	Slides 3 to 5: three definitions. The bottom: the lowest price before a recovery, which nobody can see until it is behind them. Buying low: buying a company after it fell, because you believe it will recover; the risk is that some do not. Rebalancing	Copy the three definitions into the notes box. Work Part A alone. Work Part B alone from their own ledger.	12 min

	after a crash: your pie changed shape without you doing anything (the risky slices shrank); rebalancing means buying a little of what fell and selling a little of what held, to get back to the pie you chose in Round 5. Then slides 6 and 7: the formula, with STARTING in bold on the bottom of the fraction, and one worked example (\$1,000 to \$850 is minus 15 percent). Hand out Return Math. Part A: four practice problems with the kit's prices (two gains, two losses) and the trick question (GAS from 40 to 20 and back). Part B: my own return: starting value \$1,000, my Round 8 total from the ledger, percent return by hand and checked on the calculator; then the company that hurt me most, Round 6 price to Round 8 price. Circulate; the common error is dividing by the ending value.		
3. Trading, Round 8, then Part C in pairs	Open trading at Round 8 prices for four minutes. Say once: "If your pie changed shape in the	Rebalance, buy low, or hold; record every trade in the trade log and update the Round 8 holdings	10 min

	crash, this is the round to fix it. If you are buying low, write which company and what you expect on Entry 8 first. If you are holding, that is a decision too; write it." Circulate. At four minutes, close the trading and say it plainly: the market is frozen at Round 8 until the closing bell next class. Then Part C in pairs: three investors who started in 1870 with \$1,000 and never traded again, one all in the bank, one all in RAIL, one spread four ways; their totals at the peak (Round 6), in the crash (Round 7) and at the bottom (Round 8, today); nine percent returns and two questions.	table. Then Part C with a partner: compute the nine returns, circle the ledger they would rather have held and write why.	
4. Investor Journal, Entry 8	Prompt: what I did, why, what happened, my total in dollars, plus: my percent return so far is ____ (from Part B), and the company that hurt me most fell ____ percent from 1928 to 1932.	Write Entry 8 with the total and the two percents.	6 min
Closure	See Closure.		4 min

Totals 40 minutes: 3 + 5 + 12 + 10 + 6 + 4.

Questions to ask

Monitor understanding:

- Start \$1,000, end \$850. What goes on top of the fraction? What goes on the bottom?
- STEEL fell from 90 to 25. Is that a 65 percent drop or a 72 percent drop? How do you know?
- Your return is negative. Does that mean you did worse than everyone else?

Deepen learning:

- The all-bank ledger is up 26.8 percent after eight rounds and never dropped once. The all-RAIL ledger was up 40 percent at the peak and is down 80 percent today. Which one would you want if this were money for college in ten years? For rent next month?
- Why can a company fall 50 percent and then rise 50 percent and still not be back where it started?
- RADIO is \$3 today and was \$100 four years ago. Is \$3 cheap? What would have to be true for it to be a bargain, and what would have to be true for it to be a trap?
- The dispatch says thousands of real banks failed and the money in them is gone. Our bank never fails. What does that change about the safe line on your ledger, in the game and in real life?

Check for understanding

Step 2, Part A: clipboard yes or no per student on “divides by the starting value” and “puts a minus sign on a loss.” Part C: yes or no on “the all-RAIL ledger at the bottom is negative and the mixed ledger is not.” A “got it” answer sounds like: “Ending minus starting, over starting. Mine is 640 minus 1,000, over 1,000, so minus 36.0 percent.”

Closure (Time: 4 min)

Exit card: “Write the percent return formula in words. Then: my return today is ___ percent, and the all-bank ledger’s return today is ___ percent. One sentence: which is better, and better for what?” Three students read the last sentence aloud. Sort into “got it” (formula correct, both numbers present) and “needs more” (a worked-example card goes in the folder before Lesson 2.19, where the report needs this number).

Differentiation and supports

- ELL: the formula is shown as a picture (a fraction with “end minus start” on top and “start” on the bottom) on the slide and the handout; the words peak and bottom have a mountain drawing; sentence starter “My return is ___ percent because ___” on the journal template; pair with a partner for Part C.
- IEP and 504 (general): calculators for all; Part A can be two problems instead of four; the handout has the subtraction line and the division line printed as separate boxes; extended time by finishing Part C at the start of Lesson 2.19; a student who cannot compute Part B by hand does it with the teacher at the desk, one line at a time, from the ledger’s Round 8 total.

- Grade 6 support: the practice problems use round numbers (start \$100, end \$120); Part C is the all-bank and the all-RAIL ledgers only, with the mixed ledger done as a class.
- Grade 8 stretch: the student also computes what percent a company must climb from its 1932 price to get back to its 1928 price (GAS needs 100 percent; RADIO, from 3 to 100, needs more than 3,000 percent) and explains why it is not the same as the percent it fell; and finds the company with the biggest percent swing on their own ledger.
- UDL checkpoint used: multiple means of representation (the formula as words, as a picture, and as a worked example) and multiple means of action and expression (Part B uses the student's own numbers; Part C can be answered with a circled letter and one sentence).
- No-kitchen or no-machine alternative: not needed; the lesson is paper, a board and calculators. Without calculators, Part A is the grade 6 rows and Part B is checked by the teacher.

Assessment

Objective	Standards met	Learning task	Assessment (formative or summative; tool)
1. State and compute percent return	CDOS 3a Basic Skills; National FCS 3.3.4	Handout Part A; exit card	Formative; handout checked against the key
2. Compute own return and one company's fall from peak to bottom	IGLR 4. e); Std 3 PI B; CDOS 3a Basic Skills	Handout Part B; Entry 8	Formative now; summative on Rubric 02 criterion 4 (the report's math), where the percent return is required
3. Compare safe, one-company and mixed ledgers and decide	FCL 2. c); FCL 2. f); National FCS 3.3.4	Handout Part C and the rebalance or buy-low decision in Entry 8	Formative now; summative on Rubric 02 criterion 2 (analysis and decision making)

Homework

None. Optional: pick any price you know (a game, a sneaker, a phone) that changed this year and compute the percent change.

Connections

Math: percent change is a grade 7 standard (ratios and proportional relationships); the math teacher can grade Part A as a math assignment the same week if both teachers agree, and FACS gets students who have seen the formula twice in two rooms. Social studies: 1932 as the bottom of the Depression, with the kit's numbers (the Dow at 41, one worker in four out of work).

Sources

- NYS Middle Level CTE FACS Content Module 01, Individual Growth and Life Readiness, line 4. e).
- NYS Middle Level CTE Theme Module 03, Financial and Consumer Literacy, lines 2. c) and 2. f).
- NYS FACS Standard 3, Intermediate; CDOS 3a Basic Skills; National FCS 3.0 competency 3.3.4, all per 02 Standards/STANDARDS CROSSWALK.md.
- Handout 02.14 - The Gilded Age Market Kit.md: the price table (page 2) and the Round 8 dispatch (page 3). Every price in the handout is from the kit; the three Part C ledgers are built from it and shown in the handout's teacher key.
- My own materials: my earlier market simulation plan (the recovery and buy-low day, the return calculations day, the rebalancing day, and the standards note "Return calculations involve percentage change: $(\text{Final Value} - \text{Initial Value}) / \text{Initial Value}$ ").

Teacher notes

- If the period runs short, cut Part C to the all-bank and all-RAIL columns only and skip the mixed one. Never cut Part B; every student needs their own percent return for the report.
- My earlier plan taught risk-adjusted return (the Sharpe ratio idea). It is cut here for time and grade level; the grade 8 stretch (the climb back vs the fall) carries the idea that where you measure from matters.
- Common mistake: dividing by the ending value. The handout prints "start" on the bottom in bold; say it three times anyway.
- Common mistake: a student with a negative return thinks they "failed." Point at Part C: the all-RAIL ledger is down 80 percent today and the mixed one has given back most of its gain. The report scores the reasoning, not the rank.
- The Part C mixed ledger is up 24.5 percent at the bottom and the all-bank ledger is up 26.8 percent. Students will say the bank won. Say: it did, today, by two points, after 62 years, and the mixed ledger was up 189 percent four years ago. Then ask who would have held the mixed ledger through 1929 without selling. That is the honest version of the lesson.
- Next class is the closing bell at 1936 prices. Freeze the market at Round 8 at the end of this period; nobody trades before Lesson 2.19. Do not say what 1936 prices look like.